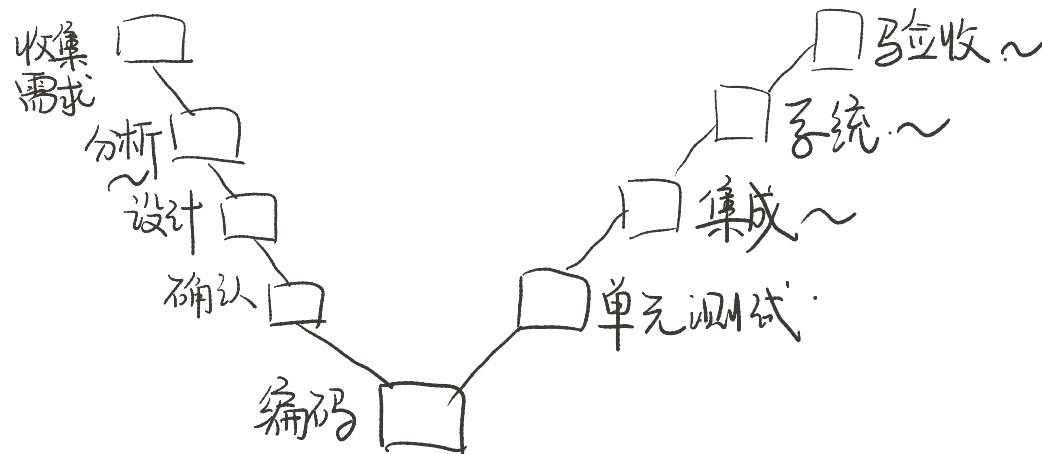
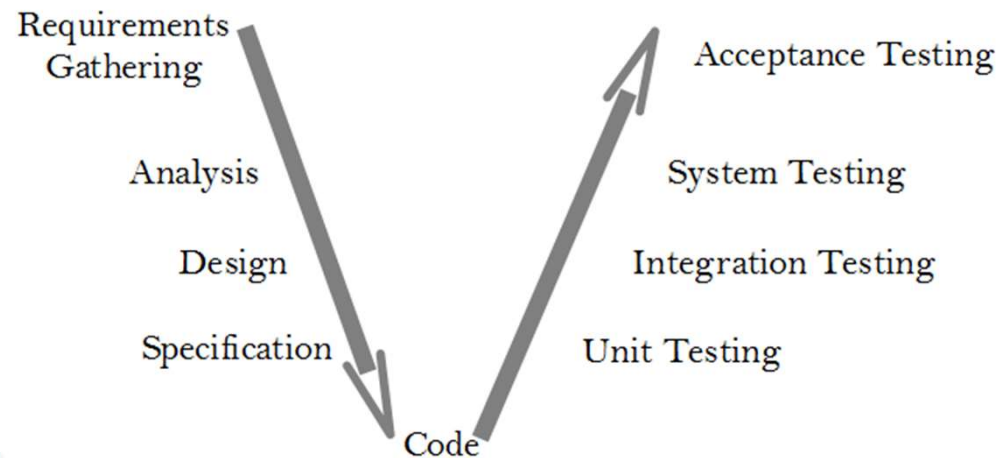


Quiz

1. Please draw the model diagram of V model and briefly describe the shortcomings of V model and how to solve the shortcomings of V model?

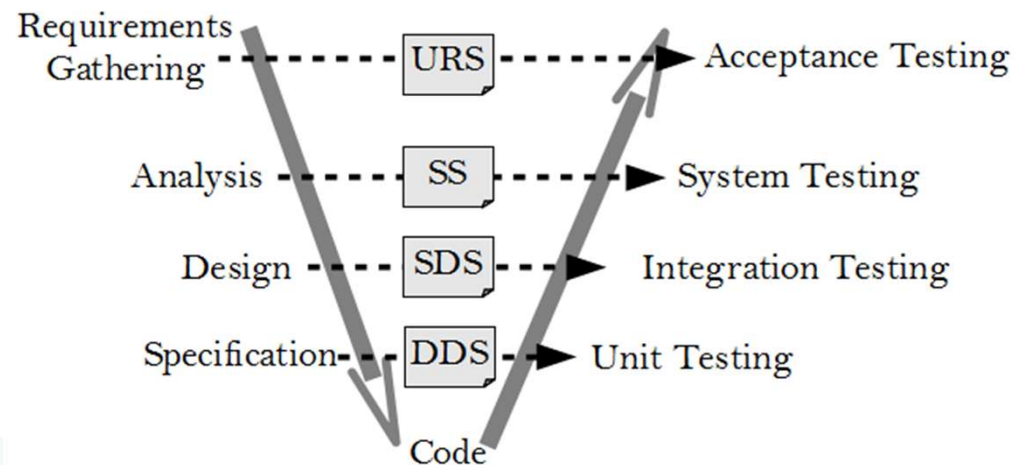


V Model



- Extension of the Waterfall model
 - emphasizes **Verification & Validation** by marking the relationships between each phase of the life cycle and testing activities
- Once the code implementation is finished the testing begins.
- Starts with unit testing, and moves up one test level at a time until the acceptance testing phase is completed

V Model



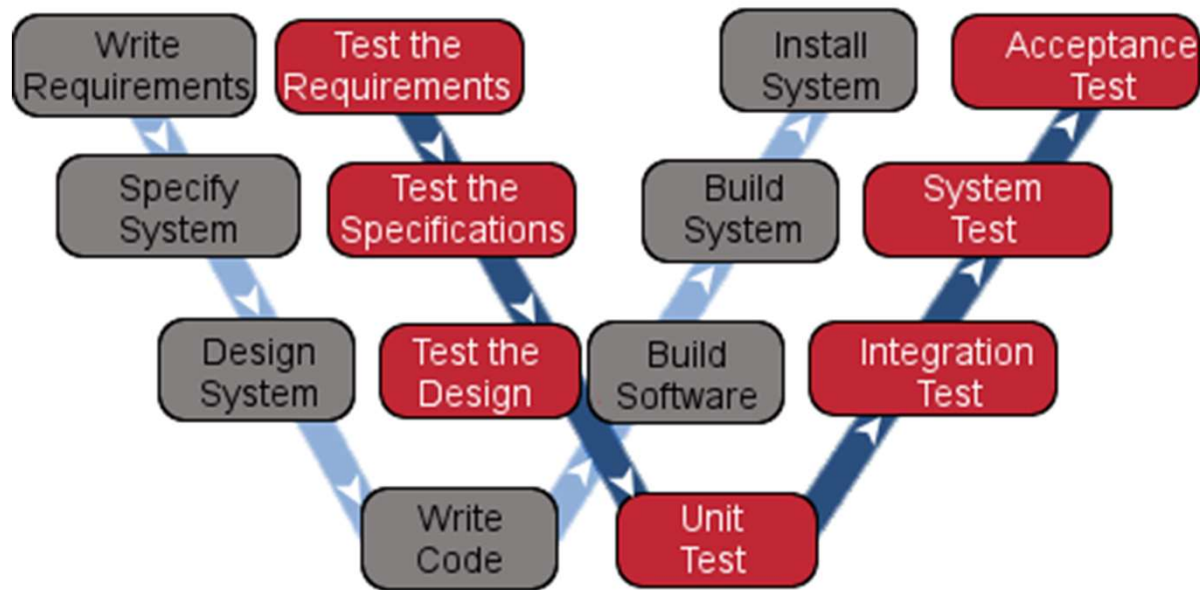
- Each document produced is associated with pairs of phases in the model.
 - (a) the User Requirements Specification. **URS**
 - (b) the System Requirements Specification, **SRS**
 - (c) the System Design Specifications, **SDS**
 - (d) Detailed Design Specifications, **DDS**

V Model – Strength& Weakness

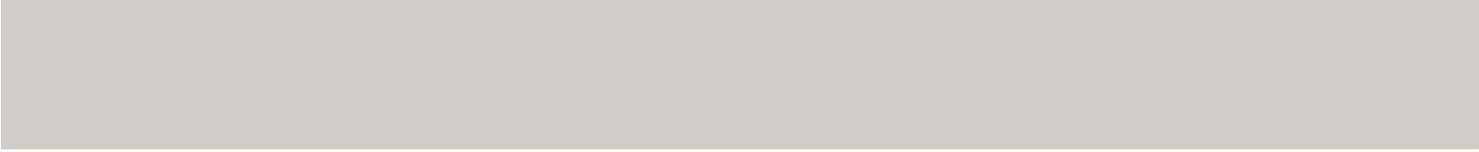
Strength	Weakness
It is simple and easy to manage due to the rigidity of the model	Like the Waterfall model , there is no working software produced until late during the life cycle
It encourages Verification and Validation at all phases	It is unsuitable where the requirements are at a moderate to high risk of changing .
Each phase has specific deliverables and a review process.	It has been suggested too that it is a poor model for long, complex and object-oriented projects
It gives equal weight to testing alongside development rather than treating it as an afterthought at the end.	

4. W Model

The W model



- Extension of V Model/**Both V**
- Testing is not after the code implementation .
- **Parallel** to the development process, the test process is carried out.
- **Co-operation** between development and testing
- **Testing** is more than just construction, execution and evaluation of test cases.

- 
1. Please design testcases based on the decision table method for the following right triangle problem with side a,b,c. (40 scores)

Side a, b, and c should be positive integers ranging from 1 to 100;

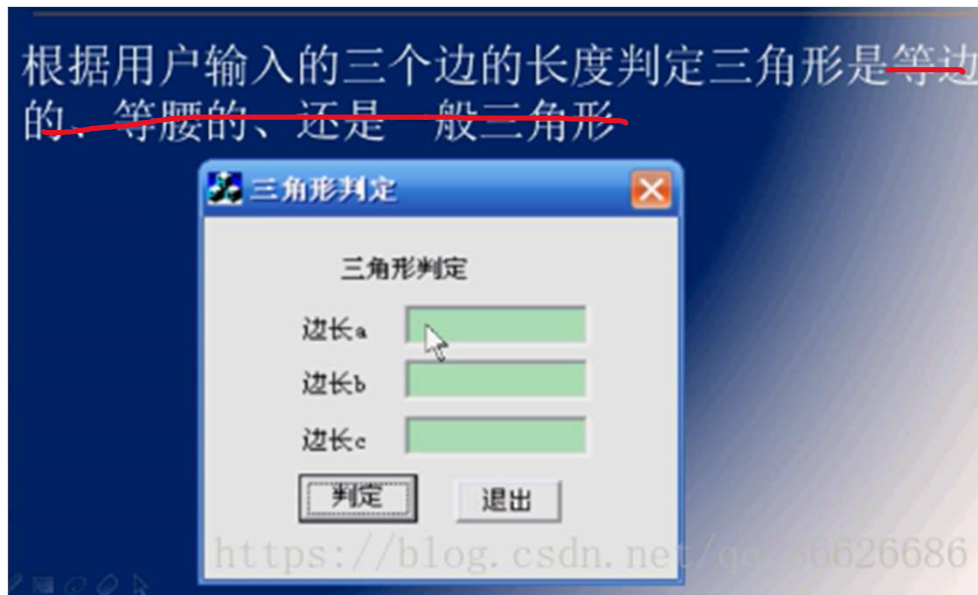
If any side is less than 1 or greater than 100, the output is “invalid input”;

If the sum of any two sides is less than or equal to the third side, the output is “non-triangular”;

If the sum of the squares of any two sides is equal to the square of the third side, the output is “a right triangle”; else the output is "general triangle";

- 1) Please draw the decision table.
 - 2) Please list the testcases.
- 

- 直角三角形的例子用判定表法进行测试用例的设计；



是否为直角三角形

- 假设三边边长a、b和c均为1到100之间的正整数，直角三角形判定需满足；
- 当三边中任意两边边长之和小于等于第三边，则输出“非三角形”
 - 当其中两边边长的平方和等于第三边的平方，则输出“直角三角形”
 - 当任意两边边长的平方和都不等于第三边的平方，则输出“一般三角形”
 - 当三边中任意一边边长小于1或者大于100，则给出“输入无效”的提示信息

编号	原因	编号	结果
c1	$1 \leq a$	e1	输入无效
c2	$a \leq 100$	e2	非三角形
c3	$1 \leq b$	e3	一般三角形
c4	$b \leq 100$	e4	直角三角形
c5	$1 \leq c$		
c6	$c \leq 100$		
c7	$a + b > c$		
c8	$a + c > b$		
c9	$b + c > a$		
c10	$a^2 + b^2 = c^2$		
c11	$a^2 + c^2 = b^2$		
c12	$b^2 + c^2 = a^2$		

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
条件桩	c1	F						T	T	T	T	T	T	T	T	T	T	T
	c2		F					T	T	T	T	T	T	T	T	T	T	T
	c3			F				T	T	T	T	T	T	T	T	T	T	T
	c4				F			T	T	T	T	T	T	T	T	T	T	T
	c5					F		T	T	T	T	T	T	T	T	T	T	T
	c6						F	T	T	T	T	T	T	T	T	T	T	T
	c7							F	T	T	T	T	T	T	T	T	T	T
	c8								F	T	T	T	T	T	T	T	T	T
	c9									F	T	T	T	T	T	T	T	T
	c10										F	F	F	T	T	T	F	T
	c11										F	F	T	F	T	F	T	T
	c12										F	T	F	F	F	T	T	T
动作桩	e1	x	x	x	x	x	x											
	e2							x	x	x								
	e3										x							
	e4											x	x	x				

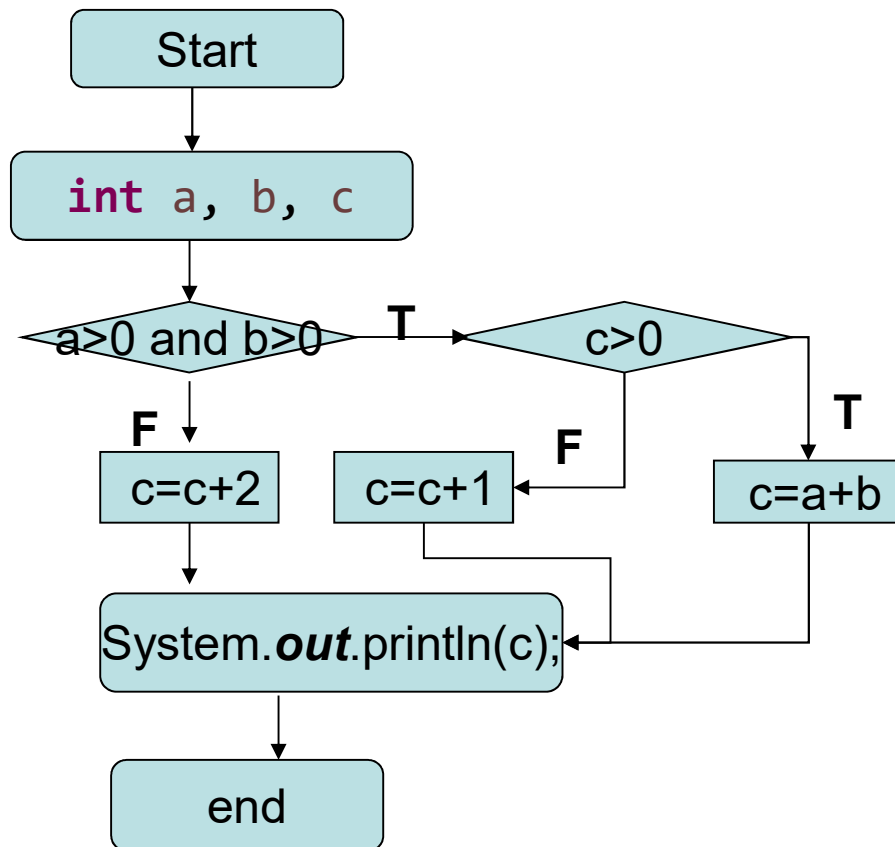
Rule	a	b	c	Expected Output
1	-1	50	50	invalid input
2	101	50	50	invalid input
3	50	-1	50	invalid input
4	50	101	50	invalid input
5	50	50	-1	invalid input
6	50	50	101	invalid input
7	50	1	60	non-triangular
8	50	60	1	non-triangular
9	60	50	1	non-triangular
10	40	50	60	general triangle
11	30	40	50	right triangle
12	30	50	40	right triangle
13	50	30	40	right triangle

2. Please complete the testcases design using the following white-box testing techniques. (50 scores)

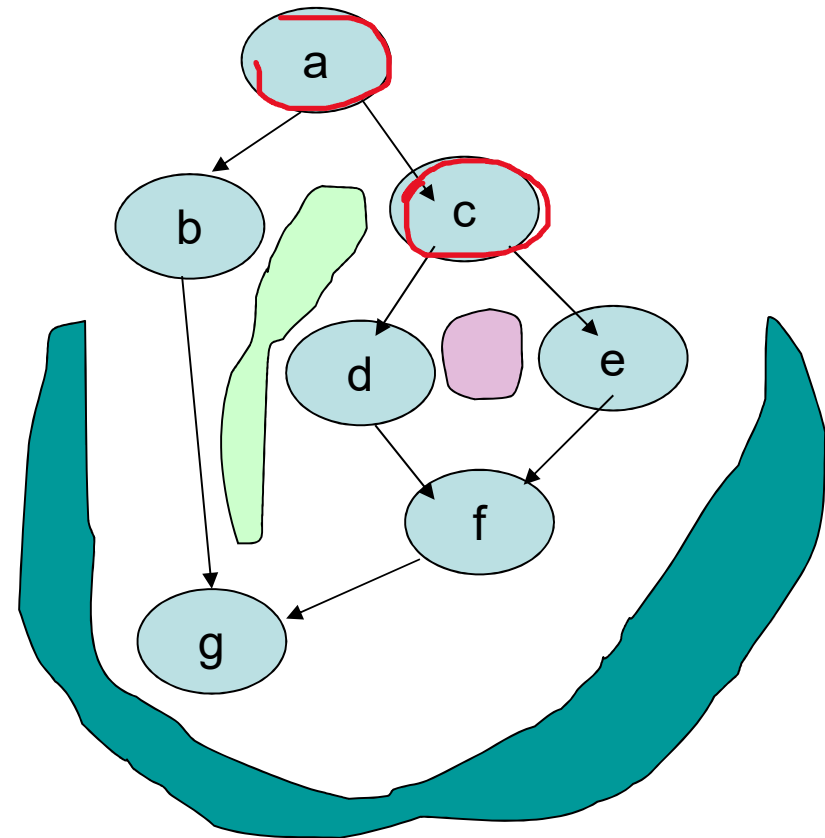
- 1) Please draw the control flow chart.
- 2) Please list all the decisions and their conditions branches.
- 3) Please use the condition combination testing method to design testcases.
- 4) Please use the path coverage method to design testcases.
- 5) Please use the basis path testing method to design testcases.

```
1  public static void showC() {  
2      int a, b, c ;  
3      if (a > 0 & b > 0)  
4      {  
5          if (c > 0) {  
6              c = a + b;  
7          }  
8          else  
9              c = c + 1;  
10     }  
11     else  
12         c = c + 2;  
13     System.out.println(c);  
14 }
```

Program flow graph



Control flow graph



答案

- 判定2个:

- $a > 0$ and $b > 0$
- $c > 0$

- 判定分支4个:

- $a > 0$ and $b > 0$, **M**;
- $a \leq 0$ or $b \leq 0$, **-M**
- $c > 0$, **N**;
- $c \leq 0$, **-N**

- 条件分支,6个:

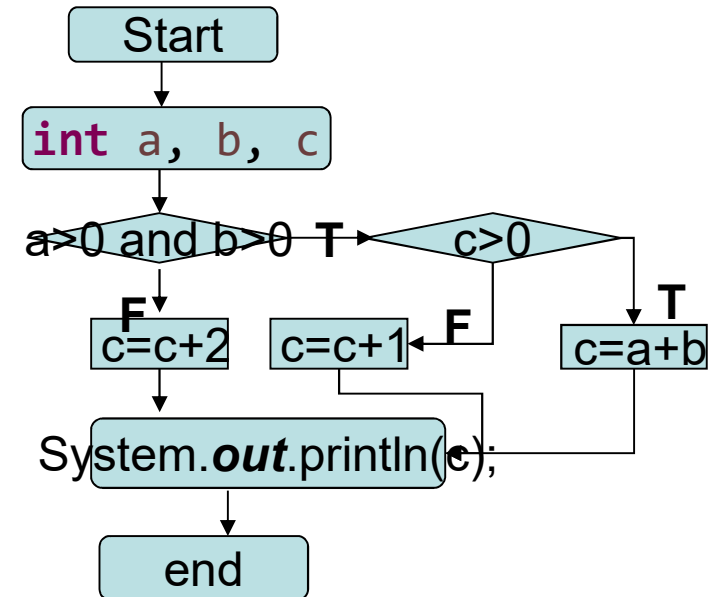
- $a > 0$, **X1**;
- $a \leq 0$, **-X1**;
- $b > 0$, **X2**;
- $b \leq 0$, **-X2**;
- $c > 0$, **X3**;
- $c \leq 0$, **-X3**

1. 判定一条件覆盖法:

要覆盖所有的判定分支与条件分支

覆盖判定:3个用例,检查未覆盖的条件,补充

- $a > 0$ and $b > 0$, M;
 $a \leq 0$ or $b \leq 0$, -M
 $c > 0$, N;
 $c \leq 0$, -N
- $a > 0$, X1;
 - $a \leq 0$, -X1;
 - $b > 0$, X2;
 - $b \leq 0$, -X2;
 - $c > 0$, X3;
 - $c \leq 0$, -X3



测试用例	a, b, c	执行路径	覆盖条件	覆盖判定	预期结果
1	-1,-1,1	1-2-3-7-8	-X1, -X2	-M	c=3
2	1,1,-1	1-2-3-4-6-8	X1, X2, -X3	M,-N	c=0
3	1,1,1	1-2-3-4-5-8	X1, X2, X3	M,N	c=2

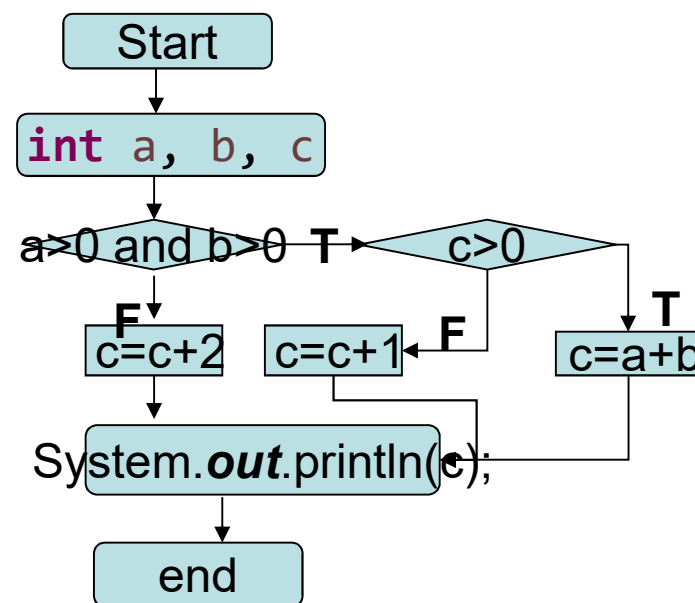
2. 条件组合覆盖

1. $a > 0, b > 0$ TT
2. $a > 0, b \leq 0$ TF
3. $a \leq 0, b > 0$ FT
4. $a \leq 0, b \leq 0$ FF
5. $c > 0$ T
6. $c \leq 0$ F

2. 条件组合覆盖法:

要覆盖每个判定的所有条件组合

1. $a > 0, b > 0$ **TT**
2. $a > 0, b \leq 0$ **TF**
3. $a \leq 0, b > 0$ **FT**
4. $a \leq 0, b \leq 0$ **FF**
5. $c > 0$ **T**
6. $c \leq 0$ **F**

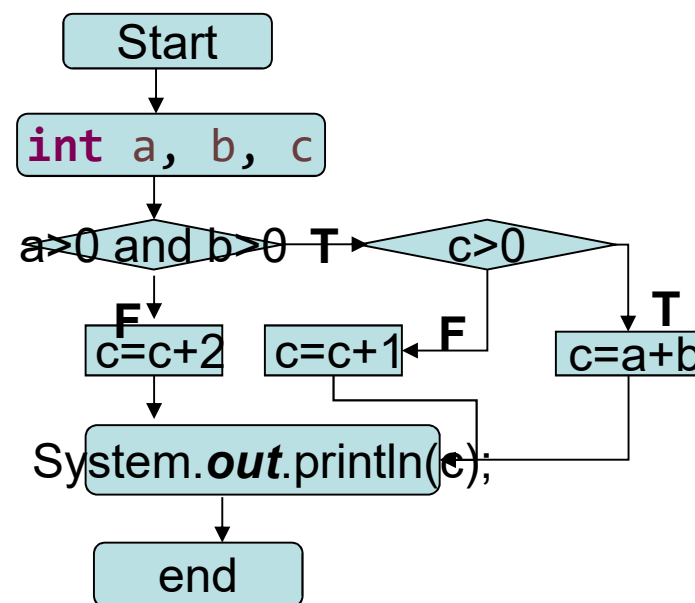


测试用例	a, b, c	执行路径	覆盖的条件组合	预期结果
1	1,1,1	1-2-3-4-5-8	1,5	c=2
2	1,-1,-1	1-2-3-7-8	2	c=1
3	-1,1,1	1-2-3-7-8	3	c=3
4	-1,-1,-1	1-2-3-7-8	4	c=1
5	1,1,-1	1-2-3-4-6-8	1,6	c=0

2. 条件组合覆盖法:

要覆盖每个判定的所有条件组合

1. $a > 0, b > 0$ **TT**
2. $a > 0, b \leq 0$ **TF**
3. $a \leq 0, b > 0$ **FT**
4. $a \leq 0, b \leq 0$ **FF**
5. $c > 0$ **T**
6. $c \leq 0$ **F**

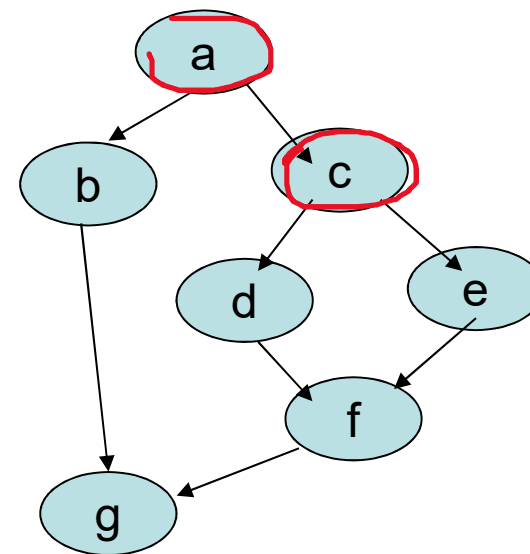


测试用例	a, b, c	执行路径	覆盖的条件组合	预期结果
1	1,1,1	1-2-3-4-5-8	1,5	c=2
2	1,-1,-1	1-2-3-7-8	2	c=1
3	-1,1,1	1-2-3-7-8	3	c=3
4	-1,-1,-1	1-2-3-7-8	4	c=1
5	1,1,-1	1-2-3-4-6-8	1,6	c=0

3. 路径覆盖法

- $a.(b+c.(d+e).f).g$
- $=abg+acdfg+acefg$

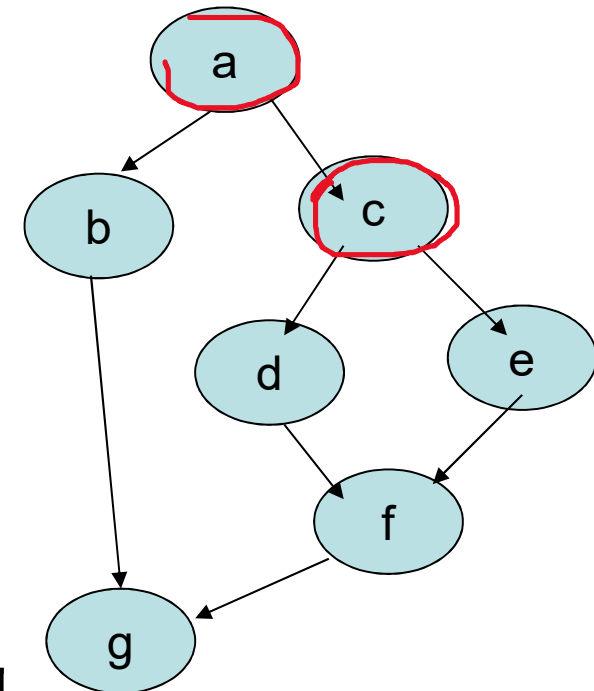
测试用例	a, b, c	执行路径	预期结果
1	1,1,1	acefg	c=2
2	1,1,-1	acdfg	c=0
3	-1,1,1	abg	c=3



对比一下几种覆盖的效果：未必覆盖条件组合，条件

4. 基本路径覆盖法

- 基本路径数:
- 3个区域
- 2个判定+1
- $8-7+2=3$
- 基本路径集:
- $abg/acdfg/acefg$



测试用例	a, b, c	执行路径	预期结果
1	1,1,1	acefg	c=2
2	1,1,-1	acdfg	c=1
3	-1,1,1	abg	c=3